

60 Easy Shell Scripting Programs

Bash / Linux Practical & Interview Practice

How to run: save as **file.sh**, then use **chmod +x file.sh** and **./file.sh**.

1. Hello World

```
#!/bin/bash
echo "Hello World"
```

2. Print Name

```
#!/bin/bash
read -p "Enter your name: " name
echo "Hello $name"
```

3. Add Two Numbers

```
#!/bin/bash
read -p "Enter first number: " a
read -p "Enter second number: " b

sum=$((a+b))
echo "Sum = $sum"
```

4. Even or Odd

```
#!/bin/bash
read -p "Enter number: " n

if [[ $(($n%2)) -eq 0 ]]
then
    echo "Even"
else
    echo "Odd"
fi
```

5. Positive, Negative or Zero

```
#!/bin/bash
read -p "Enter number: " n

if [[ $n -gt 0 ]]
then
    echo "Positive"
elif [[ $n -lt 0 ]]
then
    echo "Negative"
else
    echo "Zero"
fi
```

6. Greatest of Two Numbers

```
#!/bin/bash
read -p "Enter first number: " a
read -p "Enter second number: " b

if [[ $a -gt $b ]]
then
    echo "$a is greatest"
else
```

```
        echo "$b is greatest"
    fi
```

7. Greatest of Three Numbers

```
#!/bin/bash
read -p "Enter three numbers: " a b c

if [[ $a -gt $b && $a -gt $c ]]
then
    echo "$a is greatest"
elif [[ $b -gt $c ]]
then
    echo "$b is greatest"
else
    echo "$c is greatest"
fi
```

8. Print 1 to 10

```
#!/bin/bash

for ((i=1; i<=10; i++))
do
    echo $i
done
```

9. Print Even Numbers 1 to 20

```
#!/bin/bash

for ((i=1; i<=20; i++))
do
    if [[ $((i%2)) -eq 0 ]]
    then
        echo $i
    fi
done
```

10. Sum of N Numbers

```
#!/bin/bash
read -p "Enter n: " n

sum=0

for ((i=1; i<=n; i++))
do
    sum=$((sum+i))
done

echo "Sum = $sum"
```

11. Multiplication Table

```
#!/bin/bash
read -p "Enter number: " n

for ((i=1; i<=10; i++))
do
    echo "$n x $i = $((n*i))"
done
```

12. Factorial

```
#!/bin/bash
```

```

read -p "Enter number: " n

fact=1

for ((i=1; i<=n; i++))
do
    fact=$((fact*i))
done

echo "Factorial = $fact"

```

13. Reverse a Number

```

#!/bin/bash
read -p "Enter number: " n

rev=0

while [[ $n -gt 0 ]]
do
    r=$((n%10))
    rev=$((rev*10+r))
    n=$((n/10))
done

echo "Reverse = $rev"

```

14. Palindrome Number

```

#!/bin/bash
read -p "Enter number: " n

original=$n
rev=0

while [[ $n -gt 0 ]]
do
    r=$((n%10))
    rev=$((rev*10+r))
    n=$((n/10))
done

if [[ $original -eq $rev ]]
then
    echo "Palindrome"
else
    echo "Not Palindrome"
fi

```

15. Check File Exists

```

#!/bin/bash
read -p "Enter filename: " file

if [[ -f $file ]]
then
    echo "File exists"
else
    echo "File does not exist"
fi

```

16. Check Directory Exists

```

#!/bin/bash
read -p "Enter directory name: " dir

if [[ -d $dir ]]

```

```

then
    echo "Directory exists"
else
    echo "Directory does not exist"
fi

```

17. Count Lines in a File

```

#!/bin/bash
read -p "Enter filename: " file

wc -l $file

```

18. Search Word in File

```

#!/bin/bash
read -p "Enter word: " word
read -p "Enter filename: " file

grep "$word" "$file"

```

19. Create Backup

```

#!/bin/bash
read -p "Enter filename: " file

cp "$file" "$file.bak"

echo "Backup created"

```

20. Simple Menu

```

#!/bin/bash

echo "1. Date"
echo "2. Calendar"
echo "3. Current Directory"
echo "4. List Files"

read -p "Enter choice: " ch

case $ch in
    1) date ;;
    2) cal ;;
    3) pwd ;;
    4) ls ;;
    *) echo "Invalid choice" ;;
esac

```

21. Check Vowel or Consonant

```

#!/bin/bash

read -p "Enter a character: " ch

if [[ $ch == "a" || $ch == "e" || $ch == "i" || $ch == "o" || $ch == "u" ]]
then
    echo "Vowel"
else
    echo "Consonant"
fi

```

22. Check Character is Digit

```

#!/bin/bash

```

```

read -p "Enter character: " ch

if [[ $ch =~ [0-9] ]]
then
    echo "Digit"
else
    echo "Not a digit"
fi

```

23. Check Leap Year

```

#!/bin/bash

read -p "Enter year: " year

if [[ $((year%400)) -eq 0 || $((year%4)) -eq 0 && $((year%100)) -ne 0 ]]
then
    echo "Leap year"
else
    echo "Not a leap year"
fi

```

24. Swap Two Numbers

```

#!/bin/bash

read -p "Enter a: " a
read -p "Enter b: " b

temp=$a
a=$b
b=$temp

echo "a = $a"
echo "b = $b"

```

25. Simple Calculator

```

#!/bin/bash

read -p "Enter first number: " a
read -p "Enter second number: " b
read -p "Enter operator: " op

case $op in
    +) echo $((a+b)) ;;
    -) echo $((a-b)) ;;
    \*) echo $((a*b)) ;;
    /) echo $((a/b)) ;;
    *) echo "Invalid operator" ;;
esac

```

26. Print 10 to 1

```

#!/bin/bash

for ((i=10; i>=1; i--))
do
    echo $i
done

```

27. Sum of Even Numbers

```

#!/bin/bash

read -p "Enter n: " n

```

```

sum=0

for ((i=2; i<=n; i+=2))
do
    sum=$((sum+i))
done

echo "Sum = $sum"

```

28. Sum of Odd Numbers

```

#!/bin/bash

read -p "Enter n: " n
sum=0

for ((i=1; i<=n; i+=2))
do
    sum=$((sum+i))
done

echo "Sum = $sum"

```

29. Count Digits

```

#!/bin/bash

read -p "Enter number: " n
count=0

while [[ $n -gt 0 ]]
do
    n=$((n/10))
    count=$((count+1))
done

echo "Digits = $count"

```

30. Sum of Digits

```

#!/bin/bash

read -p "Enter number: " n
sum=0

while [[ $n -gt 0 ]]
do
    r=$((n%10))
    sum=$((sum+r))
    n=$((n/10))
done

echo "Sum = $sum"

```

31. Product of Digits

```

#!/bin/bash

read -p "Enter number: " n
product=1

while [[ $n -gt 0 ]]
do
    r=$((n%10))
    product=$((product*r))
    n=$((n/10))
done

```

```
echo "Product = $product"
```

32. Check Prime Number

```
#!/bin/bash

read -p "Enter number: " n
count=0

for ((i=1; i<=n; i++))
do
    if [[ $(($n%i)) -eq 0 ]]
    then
        count=$((count+1))
    fi
done

if [[ $count -eq 2 ]]
then
    echo "Prime"
else
    echo "Not Prime"
fi
```

33. Print Prime Numbers

```
#!/bin/bash

read -p "Enter n: " n

for ((num=2; num<=n; num++))
do
    count=0

    for ((i=1; i<=num; i++))
    do
        if [[ $(($num%i)) -eq 0 ]]
        then
            count=$((count+1))
        fi
    done

    if [[ $count -eq 2 ]]
    then
        echo $num
    fi
done
```

34. Fibonacci Series

```
#!/bin/bash

read -p "Enter n: " n

a=0
b=1

for ((i=1; i<=n; i++))
do
    echo -n "$a "
    c=$((a+b))
    a=$b
    b=$c
done
```

35. Find Largest Digit

```
#!/bin/bash

read -p "Enter number: " n

max=0

while [[ $n -gt 0 ]]
do
    r=$((n%10))

    if [[ $r -gt $max ]]
    then
        max=$r
    fi

    n=$((n/10))
done

echo "Largest digit = $max"
```

36. Star Pattern

```
#!/bin/bash

for ((i=1; i<=5; i++))
do
    for ((j=1; j<=i; j++))
    do
        echo -n "* "
    done
    echo
done
```

37. Reverse Star Pattern

```
#!/bin/bash

for ((i=5; i>=1; i--))
do
    for ((j=1; j<=i; j++))
    do
        echo -n "* "
    done
    echo
done
```

38. Number Pattern

```
#!/bin/bash

for ((i=1; i<=5; i++))
do
    for ((j=1; j<=i; j++))
    do
        echo -n "$j "
    done
    echo
done
```

39. Find String Length

```
#!/bin/bash

read -p "Enter string: " str

echo "Length = ${#str}"
```

40. Reverse String

```
#!/bin/bash

read -p "Enter string: " str

echo "$str" | rev
```

41. Check String Palindrome

```
#!/bin/bash

read -p "Enter string: " str

rev=$(echo "$str" | rev)

if [[ "$str" == "$rev" ]]
then
    echo "Palindrome"
else
    echo "Not Palindrome"
fi
```

42. Count Vowels

```
#!/bin/bash

read -p "Enter string: " str

echo "$str" | grep -o -i '[aeiou]' | wc -l
```

43. Convert Lowercase to Uppercase

```
#!/bin/bash

read -p "Enter string: " str

echo "$str" | tr 'a-z' 'A-Z'
```

44. Convert Uppercase to Lowercase

```
#!/bin/bash

read -p "Enter string: " str

echo "$str" | tr 'A-Z' 'a-z'
```

45. Check File Permission

```
#!/bin/bash

read -p "Enter filename: " file

if [[ -r $file ]]
then
    echo "Readable"
fi

if [[ -w $file ]]
then
    echo "Writable"
fi

if [[ -x $file ]]
then
    echo "Executable"
```

```
fi
```

46. Count Words in File

```
#!/bin/bash

read -p "Enter filename: " file

wc -w "$file"
```

47. Count Characters in File

```
#!/bin/bash

read -p "Enter filename: " file

wc -m "$file"
```

48. Count Lines, Words and Characters

```
#!/bin/bash

read -p "Enter filename: " file

wc "$file"
```

49. Display First 5 Lines

```
#!/bin/bash

read -p "Enter filename: " file

head -5 "$file"
```

50. Display Last 5 Lines

```
#!/bin/bash

read -p "Enter filename: " file

tail -5 "$file"
```

51. Show Current Date

```
#!/bin/bash

date
```

52. Show Current User

```
#!/bin/bash

whoami
```

53. Show Current Directory

```
#!/bin/bash

pwd
```

54. Show Logged-in Users

```
#!/bin/bash

who
```

55. Show Disk Usage

```
#!/bin/bash

df -h
```

56. Show Memory Usage

```
#!/bin/bash

free -h
```

57. Show Running Processes

```
#!/bin/bash

ps
```

58. Create Directory

```
#!/bin/bash

read -p "Enter directory name: " dir

mkdir "$dir"

echo "Directory created"
```

59. Delete File

```
#!/bin/bash

read -p "Enter filename: " file

rm "$file"

echo "File deleted"
```

60. Copy File

```
#!/bin/bash

read -p "Enter source file: " source
read -p "Enter destination: " dest

cp "$source" "$dest"

echo "File copied"
```